

Part C - Question 7: Data Cleaning Report

1. Which Imputation Strategy Was Most Effective?

- **MICE Algorithm (IterativeImputer):** Most effective for numerical data (Age, BMI, Glucose, Cholesterol). It predicts missing values using relationships between columns instead of using a simple average.
- **Most Frequent Imputation:** Most effective for categorical data (Gender, Region) by filling missing values with the most common category.

2. Which Outlier Handling Method Preserved Data Quality Best?

- **Winsorization & Percentile Capping:** Preserved data quality best. Unlike Z-score and IQR methods which delete rows and reduce dataset size, capping limits extreme outlier values without deleting any patient records.

3. How Data Cleaning Improved Dataset Usability?

- **No Missing Values:** Machine learning models can now run smoothly without NaN errors.
- **No Extreme Noise:** Capping extreme outliers prevents model bias and stabilizes training.
- **Ready for ML Modeling:** The clean dataset improves model accuracy for predicting heart disease risk.